Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. Each experiment was repeated 3 times with different random seeds. All experiments were run on a single GPU with 12 GB of memory.

2 Results

Figure 1 shows the f1 score by dataset.

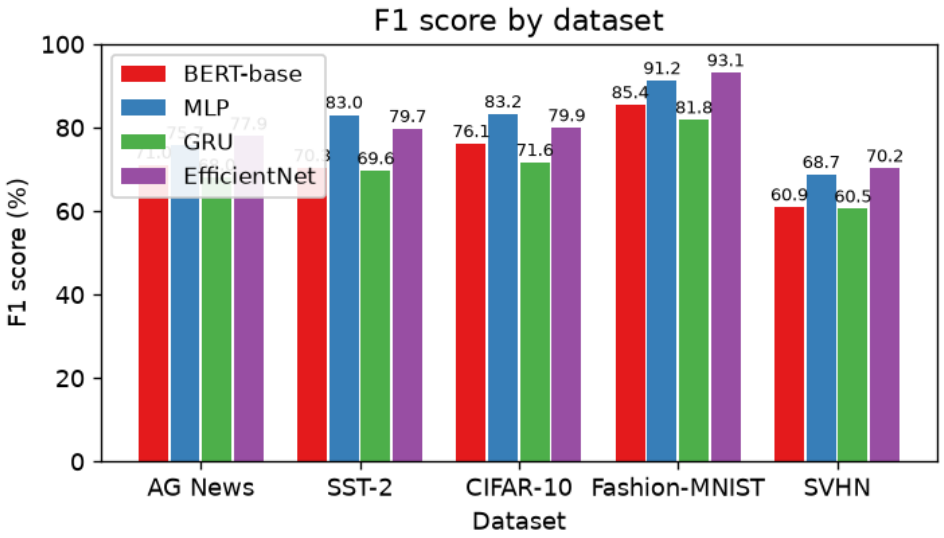


Figure 1: F1 score by dataset for the models in this experiment.

GRU achieves 69.6% F1 score on SST-2. GRU reaches 60.5% F1 score on SVHN (Figure 1). EfficientNet outperforms GRU by 10 percentage points on SVHN. On Fashion-MNIST, BERT-base obtains a F1 score of about 90% (Figure 1).

Figure 2 shows the training loss.

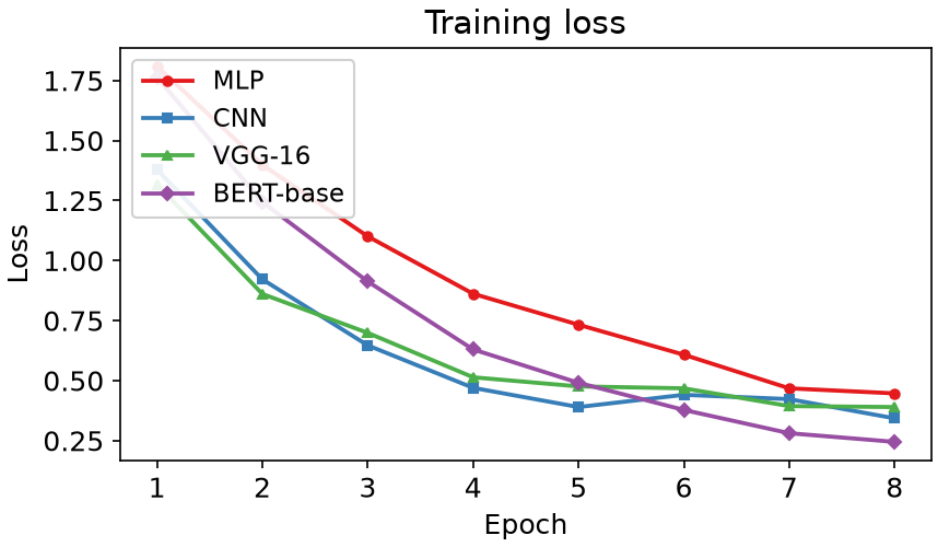


Figure 2: Training loss for the models in this experiment.

At epoch 5, the loss of MLP is 1.594 (Figure 2). CNN has a loss of 0.39 at epoch 5. As shown in Figure 2, VGG-16 has a loss of 0.600 at epoch 4.

3 Discussion

The differences between the models are discussed in the next section.